

Problem Statement

A leading retail company aims to better understand customer shopping behavior to improve sales performance, customer satisfaction, and long-term customer loyalty. The company has observed variations in purchasing patterns across different customer demographics, product categories, seasons, and shopping preferences.

Management wants to identify the key factors that influence customer purchasing decisions, including discounts, product reviews, subscription status, shipping preferences, and previous purchase behavior. Understanding these factors will help the company develop targeted marketing strategies, improve customer engagement, optimize product offerings, and increase revenue.

This project analyzes customer shopping data to uncover meaningful trends, identify valuable customer segments, evaluate the effectiveness of discounts and subscriptions, and provide data-driven recommendations for business growth.

Business Question

How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Project Objectives

- Analyze customer purchasing behavior and spending patterns.
- Identify high-value customer segments based on purchase history.
- Evaluate the impact of discounts on customer purchases.
- Compare purchasing behavior between subscribers and non-subscribers.
- Identify top-performing products and categories.
- Understand revenue contribution across different age groups and genders.
- Develop actionable business recommendations using data insights.

Deliverables

1. Data Preparation & Modeling (Python)

- Clean and transform raw customer shopping data.
- Handle missing values and data inconsistencies.
- Create derived features for analysis.
- Load cleaned data into PostgreSQL.

2. Data Analysis (SQL)

- Perform business-focused SQL analysis.
- Segment customers based on purchase history.
- Analyze revenue, loyalty, discounts, and product performance.
- Extract insights to support business decisions.

3. Visualization & Insights (Power BI)

- Build an interactive dashboard.
- Visualize customer demographics, revenue trends, product performance, and customer segments.
- Enable stakeholders to explore insights through filters and KPIs.

4. Report & Presentation

- Summarize key findings and business insights.
- Provide recommendations to improve sales and customer retention.
- Present results in a clear and visually appealing format.

5. GitHub Repository

- Store Python scripts, SQL queries, Power BI files, and project documentation in a structured repository.
- Maintain version control and project transparency.